Solving Two-Step Equations

Maths · Year 8 · Algebra

Name: _____ Class: _____ Date: _____

Goal: solve equations like 3x + 5 = 20 by undoing each step, and check your answer by substituting it back in.

1 · Starter — what do you remember?

Solve these in the space provided. No calculator.

a) x + 7 = 15 = _____

b) 3x = 21 = _____

c) x − 4 = 9 = _____

d) Simplify 4a + 3a − 2 = _____

e) If y = 6, find 2y + 3 = _____

2 · Remember (fill in as your teacher explains)

The Golden Rule: whatever you do to one side of the equation, you must do to the _____ side.

Inverse operations: + undoes ____ − undoes ____ × undoes ____ ÷ undoes ____

Undo the _____ thing that was done to x first (think: socks and shoes).

3 · Worked example — copy this down

Solve 3x + 5 = 20

− 5 from both sides

3x = 15

÷ 3 on both sides

x = 5

Check: 3 × 5 + 5 = 15 + 5 = 20 ✓

Always finish by checking — substitute your answer back in and make sure both sides are equal.

4 · Your turn — show every step

Start at whichever level you like, then work down and across. Show one step per line and check each answer. Everyone tries at least one Stretch question.

WARM-UP

1) $2x + 1 = 9$

2) $3x + 2 = 14$

3) $4x - 3 = 9$

CORE

4) $5x + 7 = 42$

5) $10 = 3x + 1$

6) $x/2 + 3 = 7$

STRETCH

7) $7 = 5x - 8$

8) $x/3 - 2 = 4$

9) I think of a number, multiply it by 4 and subtract 5. The answer is 23. What was my number?

5 · Challenge — spot the mistake

Aisha solves $5x + 3 = 23$ like this:

$5x + 3 = 23$

$2x = 20$

$x = 10$

What did Aisha do wrong? Write the correct solution and prove it works.

How well do I understand today's topic?

Colour the circle that matches how you feel: ● **Red** (need more help) ● **Amber** (nearly there) ● **Green** (confident)

One thing I could explain to someone else: _____

Answers are on the lesson slides so you can mark your own work. Curriculum alignment: Oak National Academy knowledge graphs (OGL v3.0), Oak's representation of the National Curriculum for England (2014) — not an official DfE publication.